

Reinforced Magnetic-Responsive Electro-Ionic Artificial Muscles by 3D Laser-Induced Graphene Nano-Heterostructures

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Efficient ion transport and enriched responsive modals via modulating electrochemical properties of conductivity and capacitance are essential for soft electro-ionic actuators. However, cost-effective and straightforward approaches to achieve expedited fabrication of active electrode materials capable of multimodal-responsiveness remain limited. Herein, this work reports the one-step ultrafast laser direct patterning method, to readily synthesize electro- and magneto-active electrode material, derived from the unique cobalt-phosphorus co-doped core-shell heterostructures within 3D graphene frameworks, for fulfilling the dual-mode responsive electro-ionic actuators. The designed nanofiber-structured heterointerfaces across electrodes and electrolytes further promote highly efficient electron/ion transfer. The developed soft actuator exhibits superior actuation performance of peak-to-peak displacement to 13.08 mm under an ultra-low ± 0.5 V, with doubled direct current deflection under 200 mT at 1 V, an ultrafast response of 1.38 s and long-term stability ($>90\%$ retention for $\approx 106\,000$ cycles), even detectable bending to $\approx 280\ \mu\text{m}$ under exceptional ± 10 mV. The promising demonstration of promoting differentiation and proliferation of stem cells under mechanical strain and electrical stimuli, sheds more light as well on the possibility of facilitating biomedical soft robotics with ultrahigh actuation performance.

Electro-ionic soft actuators, characterized by the lightweight, superior flexibility, high stroke, and ultralow working voltage, have emerged as promising candidates for artificial muscles, and demonstrate considerable potential in miniaturized robotics, functional structures, and biomedical science.^[6–8] Notably, the active electrode material is responsible for the high-performance of soft actuators by regulating electron/ion transport; however, challenges and limitations persist in achieving a high specific capacitance alongside robust electrical conductivity.^[9] Significant advancements have been made in the exploration and tailored design of active electrode nanomaterials, highlighted by carbon/MXene-based materials (e.g., carbon nanotubes, MXenes, and graphene) renowned for conductivity ($\approx 20\,000\ \text{S cm}^{-1}$),^[10–13] as well as hierarchical architecture-rich organic frameworks (e.g., covalent and metal-organic frameworks) for exceptional capacitance ($\approx 247\ \text{F g}^{-1}$).^[14–17] However, seamlessly integrating the fundamental advantages of both for developing high-performance active electrode nanocomposites in a straightforward and efficient

method via cost-effective procedures is challenging. Further, there is an inherent difficulty for soft actuators in achieving improvements in the blocking force (≈ 4.5 mN, 2.0 V) and actuation strain ($\approx 0.68\%$, 1.0 V) mechanically,^[18,19] and the following demonstrations still fall considerably short of practical applicability.

Multi-modal actuation strategies have been adopted to facilitate the seamless expansion of electro-ionic soft actuators into complex and dynamic contexts.^[20] Especially, in comparison to photoactive actuation inevitably suffered by its limited penetrability, the incorporation of magnetoactive properties facilitates the stimulation transmission, particularly in scenarios characterized by limited accessibility. Magnetically active materials have demonstrated considerable potential for achieving targeted delivery of therapeutics via miniaturized soft robotics requiring minimal interventional procedures.^[21,22] However, the high remanence of hard ferromagnetic materials tends to induce spontaneous self-agglomeration, rendering the fabrication process

1. Introduction

Artificial muscles, designed to emulate the characteristics of natural muscles, manifest reversible transformations in posture or motion upon receiving tethered (e.g., electrical, fluidic) or untethered (e.g., magnetic, chemical, or temperature) external stimuli,^[1–5] for possibly surpassing the inherent capabilities of their natural counterparts in real-world applications.

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challenging. Thus, harnessing a low voltage with appropriate magnetic stimuli to offer a dual-mode responsive mechanism, will enhance the high-performance electrochemical actuation on functionality expansion and universality, opening the possibilities toward specialized biomedical applications.^[23,24] To this end, developing dual-responsive active materials with high electrical conductivity, superior electro-ionic capacity, and optimal magnetic property in a facile manner will overcome the shortcomings of conventional ferromagnetic materials in terms of conductivity and capacitance.

Recently, direct laser writing (DLW) has garnered significant interest for efficient nano- and micro-scale patterning to fabricate 3D porous structures with highly conductive few-layered laser-induced graphene (LIG).^[25] Doping of magnetic metal atoms, particularly redox-active transition metal atoms, into metallic nanoparticles via DLW not only enhances the conductivity and introduces magnetic domains but also improves the electro-ionic capacitance. The participation of cobalt central has been extensively investigated for catalysis/electrocatalysis and energy storage applications,^[26–28] and their superior reversible ion insertion/extraction enables overcoming the sole dependence on the electric double-layer capacitor (EDLC).^[29] Phosphorus as dopants, owing to their larger radius than that of carbon atoms, increases the number of active sites by altering the surface and porosity of LIGs,^[30] and the lone-pair electrons in the 3p and vacant 3d orbitals of phosphorous are anticipated to attract local charges to potentially improve the electrical conductivity of transition metal compounds.^[31] However, research on the energy-level alignment of functionally doped LIGs for fabricating dual-mode magnetically responsive electro-ionic soft actuators is still lacking.

Herein, we report a one-step ultrafast DLW approach to facilitate synthesize cobalt and phosphorus (Co/P) co-doped core-shell nanoparticles incorporated into 3D hierarchical nanoporous P-doped LIG (P-LIG) frameworks (denoted as Co(O_x/P_x)@P-LIG), featuring electro- and magneto-active electrode materials for high-performance dual-mode responsive soft ionic actuators. By regulating the doping heteroatoms to achieve desired magnetic properties, the synergistic coupling effect between the morphological structure and reduced redox-active center also significantly enhanced the electrochemical performance. In addition, electrospun nanofiber-structured heterointerfaces between electrodes and electrolytes were employed to construct nano-fibrous proton exchange pathways to further promote electron/ion transfer. The achieved properties of the heterostructure improved the comprehensive actuation performance, showing peak-to-peak displacement to 13.08 mm under ± 0.5 V, and even impressively bending up to ≈ 280 μ m under ± 10 mV, ultrafast response duration of 1.38 s and long-term stability ($>90\%$ retention) in $\approx 106\,000$ cycles. The nanoscale Co(O_x/P_x)-derived active material achieved a saturated magnetization of $9.80\text{ Am}^2\text{ kg}^{-1}$ with a small remanence of $1.01\text{ Am}^2\text{ kg}^{-1}$, inducing suitable magnetoactive actuations for cellulose-based electro-ionic soft actuators. The proof-of-the-concept demonstration for biomedical-oriented soft robotics on inducing the differentiation and proliferation of mesenchymal stem cells (MSCs), providing simultaneous mechanical strain and electrical stimuli simultaneously with the developed dual-responsive soft actuators, highlights the potential for developing an automated culture platform with a compact size,

simplified structure, biocompatible nature, and impressive performance.

2. Results and Discussion

2.1. Electron/Ion Transport Mechanism and Structure Design of Soft Actuator

By combining the pseudocapacitive properties of transition metal phosphides/oxides with the superior electrical conductivity of graphene, this study aimed to demonstrate high-efficiency Co/P co-doped capacitive-conductive transport via a core-shell active electrode material, through the laser-induced P-doped defect-rich 3D graphene frameworks. Analogous to catching fireflies with a net, electrons and ions were similarly apprehended within an electric field, facilitated by charge neutralization and ionic adsorption (Figure 1a). The modulation of the poly(amic acid) (PAA) precursor and laser scribing conditions contributed to the formation of coral-like macroscopic structures in graphene, which in turn, is advantageous for the transport and storage of electrons and ions. In the presence of an electric field, the narrow pores within the graphene-sheet and coral-like networks, by virtue of the confinement effect, ensnares sub-nanoscale ions (EMIM⁺, BF₄[−] of ionic liquid, IL).^[32] The vacancies dispersed across the pore surface promoted the establishment of stable covalent linkages between the entities, in which localized alterations in electric charge within defect-rich graphene, also readily ensnaring electrons with electrostatic interaction.^[33] Structurally, the integration of the core-shell coral-like graphitic skeleton with numerous structural unit cells coated in a ring-like manner with layers of P-doped graphene, facilitates rapid electron/ion transport within such porous configurations. The attachment of transition metal compounds enveloped by defect-laden graphene to 3D LIG frameworks significantly improved bidirectional conductivity and capacitance performance by harnessing capacitive potential of pseudocapacitive materials (Figure 1b).

Designing the heterointerface between electrolytes and electrodes in electrochemical actuators requires improving the structural compatibility and electron/ion transport,^[34,35] which remains a critical challenge. In this work, a foundational fiber deposition substrate was initially established atop the prepared bacterial cellulose-based electrolyte membrane via dip-coating in a diluted Nafion solution, thereafter by direct electrospinning and high-temperature annealing to construct homogenous K⁺-doped Nafion nanoscale frameworks upon this base, displaying modulation of the nanofiber-mat heterointerface with progressive structural changes. The process and characterization of the nanofiber interface modulation are described in detail in Figures 1c and S1 and S2 and Note S1 (Supporting Information). The proposed process not only provides nanoscopic deposition zones for the later application of flexible electrode coatings, thereby augmenting the structural interaction and robustness, but also enhances the electron/ion transport efficiency across the heterointerfaces.

Starting from the process involving laser induction of the PAA precursor to its imidization with Co/P co-doping, the physical appearance of the bare polyimide (PI) film not only transitioned from sap-green to blackish color but also rapidly developed into microporous frameworks. The resultant coral-like graphene networks with core-shell transition metal nanoparticles were

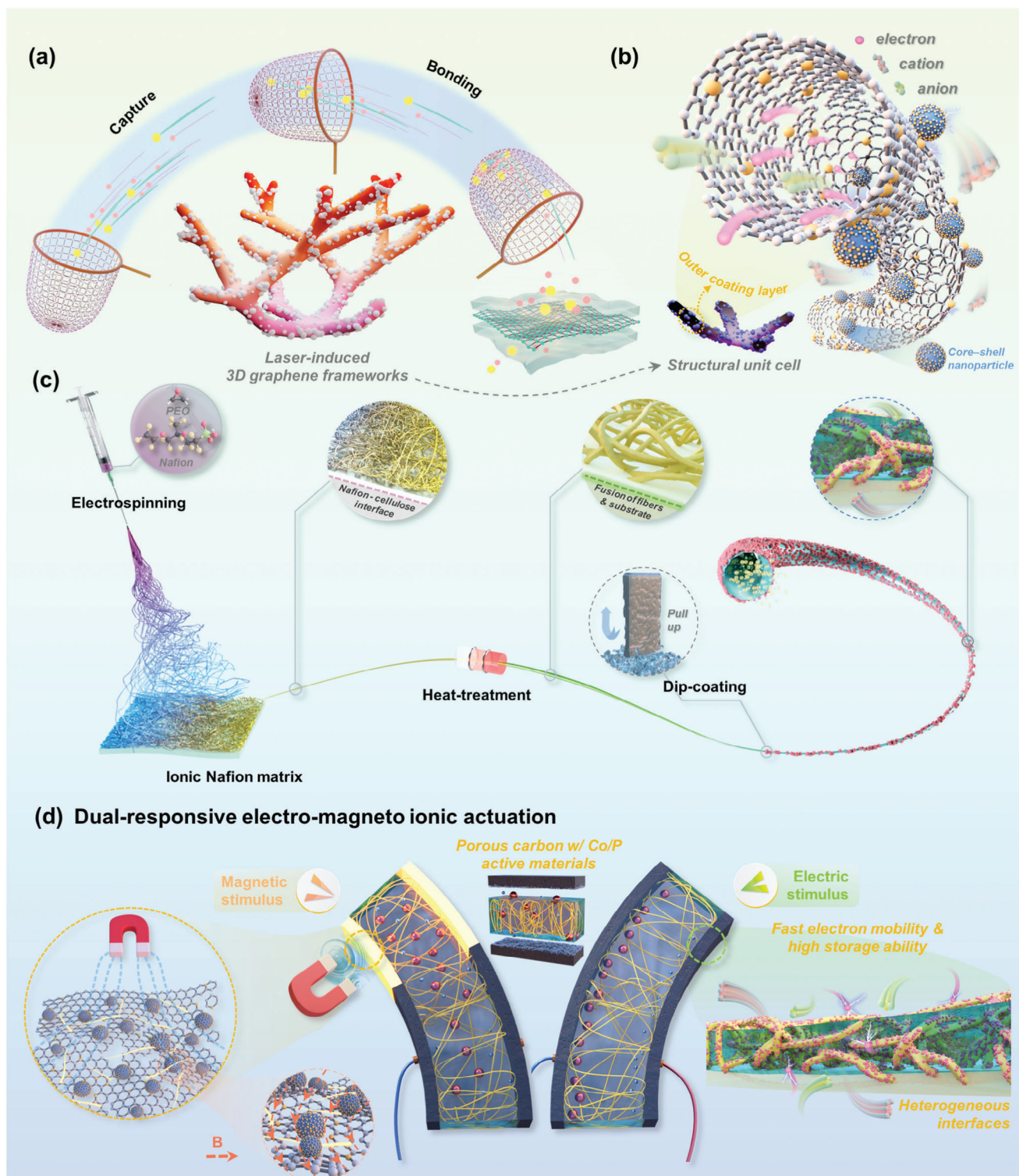


Figure 1. Schematic illustrations for the electron/ion transport mechanism of Co(O_x/P_x)-derived P-doped LIG, structure modulation, and potential application of dual-responsive ionic soft actuators. a) Coral-like core/shell structure of Co(O_x/P_x)@P-LIG capturing and bonding electrons and ions within the microscopic "net". b) Highly efficient electron/ion transport within the outer coating layer of porous carbon upon 3D graphene framework branches and between the attached core-shell Co(O_x/P_x) nanoparticles. c) Typical preparation of K⁺-doped nanofiber-structured ionic cellulose membrane through direct electrospinning and heat treatments, and the final structural outline of flexible electrodes incorporating nanoscale Co(O_x/P_x)@P-LIG with PEDOT:PSS (P.P) binder. d) Schematic representation of the dual-responsive properties of Co(O_x/P_x)@P-LIG-based soft actuator displaying the dual-modal bending actuation arisen from the electrochemical electron/ion transport and the orientation of magnetic dipoles under external magnetic field.

transformed from the organic components of doped PI, resulting in superior electron/ion transfer capability by metallic form of Co via laser-induced carbothermal reduction,^[36] other than enclosing it within and furnishing a supporting architecture of LIG. The phosphides/oxides of Co have further improved the electrochemical and electron/ion storage capacities. Furthermore, the soft ferromagnetic characteristics of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$, attributed to metallic Co and Co_2P , enabled the development of a dual-responsive actuator with exceptional electro- and magnetoactive actuation capabilities (Figure 1d). Owing to the microporous and coral-like core-shell structure, $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ facilitated the fabrication of homogeneous composite electrodes, and the incorporation of porous carbon prevented agglomeration of magnetic materials, facilitating more efficient dispersion within the electrodes.

The surface morphologies of the Co- and P-derived LIGs show the presence of coral-like porous frameworks of doped LIGs attached by transition metal nanoparticles-based core-shell structures. The structure of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ comprises a central macropore, encircled by mesopores having a coral-like graphene architecture with branches sheathed under numerous nanoparticles (Figure 2a,b; Figure S3, Supporting Information). The homogeneous precursor film with the source elements, further processed via instantaneous high-energy laser induction, enabled the homogeneous dispersion of C, O, P, and Co, as evidenced by EDS analysis (Figure S4, Supporting Information). The cross-sectional view of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ resembles that of growing coral, where the macro-porous arrangement is dense and uniform, with thickness up to 37.3 μm (Figure 2c). By regulating the weight ratio of the Co and P sources in the precursor, the morphology of the doped LIG structures can be controlled during laser induction. Specifically, lower ratios resulted in smaller pores, thicker graphene layers, and shorter structures with less uniformity, and vice versa, albeit with structurally weaker architectures (Figures S5 and S6, Supporting Information). The core-shell graphite nanosheets were further evaluated using TEM analysis (Figure 2d), showing that the few-layered LIG nanosheets are uniformly interspersed with $\text{Co}(\text{O}_x/\text{P}_x)$ nanoparticles having an average diameter of ≈ 14.76 nm. These nanoparticles are randomly separated and encapsulated by thermal decomposed LIGs (Figure 2e), which further identified by the high-resolution TEM (HRTEM) images (Figure 2f). The laser-induced nanocomposite of CoP_x core can be observed with clear stripes corresponding to (202), (210), and (201) planes of metallic Co, CoP , and Co_2P , respectively. In addition, the CoO_x core was also observed with lattice spacing of 0.21 and 0.19 nm corresponding to Co (213) and CoO (102), respectively (Figure S7, Supporting Information). The selected area electron diffraction pattern shows diffraction spots partially located on the concentric ring pattern corresponding to $\text{Co}(\text{O}_x/\text{P}_x)$ polycrystallinity.^[37] In addition, two dominant rings indexed as (100) and (002) are clearly visible, suggesting the high degree of crystallization and graphitization of polycrystalline P-doped LIG (Figure 2g).

The high-quality of the synthesized composite is confirmed by the distinct lattice patterns observed in pristine and P-doped graphene (Figure 2h,i). The interplanar spacing of C (002) plane observed in P-doped graphene slightly exceeds that of pristine LIG and is determined as ≈ 0.38 nm; this observation can be attributed to the elongated C–P bond length and larger atomic di-

mensions of the P atom in P-doped graphene.^[38] The localized thermal hotspots generated during laser induction, and the rapid heating and cooling processes create uncertainty in the crystallization process, leading to the occurrence of crystalline and amorphous heterophase states of Co/P-derived LIGs (Figure 2j). The observation of bright spots (Figure 2j-i) and diffused rings (Figure 2j-ii) in the selected-area fast Fourier transform patterns confirms the presence of heterophase, indicating the presence of coordinatively unsaturated sites, surface defects, and unpaired electrons (highlighted by blue dots); this, in turn, improves the exposure of the active sites to sub-nanoscale ions.^[39,40] At the edge of the core-shell structure (Figure 2k), nanoparticle surfaces covered with few-layered graphene having micropores can be clearly observed; this possibly contributes to the improved electron/ion adherence and transfer within the heterojunctions. High-angle annular dark-field STEM images and the corresponding elemental mapping of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ indicate the uniform presence of Co in the metallic, phosphide, and oxide states (Figure 2l–o).

The physicochemical and structural properties of nanoscale pristine LIG, P-LIG, $\text{CoO}_x\text{@LIG}$, and $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ active electrode composites were carefully investigated via spectroscopic analysis. Figure 3a shows the powder X-ray diffraction (XRD) pattern of pristine LIG presenting two intense peaks at 25.8° and 43.1° (2θ), which respectively correspond to the (002) and (100) crystallographic planes of the high graphitization of carbon.^[30] However, these crystallographic planes become to be weaker and broader with the introduction of phosphorus, since graphitic arrangement has been modified leading to the crystallinity decreases within the lattice defects arising from the variation in atomic radii between P and C atoms.^[41] By Bragg's Law, the calculated lattice spacings for pristine LIG and P-LIG samples are ≈ 0.34 and ≈ 0.37 nm, respectively, consistent with HRTEM results. Besides, $\text{CoO}_x\text{@LIG}$ exhibits one intense peak at 44.2° for metallic Co and another broader one at 32.2° for CoO .^[42] The weaker peak at 39.2° could be attributed to structure related to hydroxalite,^[43] possibly a $\text{Co(II)}\text{--Co(III)}$ layered hydroxide by atmospheric oxygen and vapor during the laser pyrolysis process. The characteristic peaks of Co_2P for $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ are distinctly visible on its (111), (002), and (300) plane, together with those of CoP , CoO , and partially reduced Co, indicating a preference for Co_2P compound formation during the laser induction, which confers the magnetoactive property to the electrode material. In addition, the variations in the interlayer spacing of graphene on the (002) plane (highlighted by brown and yellow pentagons) may suggest that $\text{Co}(\text{O}_x/\text{P}_x)$ core may doping or embedding within the graphene layers, while another intense peak could indicate localized lattice contractions or distortions, which consistent with the characteristics of a core-shell structure.

Raman analysis further demonstrates the ability of the $\text{Co}(\text{O}_x/\text{P}_x)$ core in improving electron/ion transport by introducing additional electronic states. As shown in Figure 3b, the obtained spectra are deconvoluted into four bands (D1, D3, D4, and G) using Gaussian-Lorentzian peak fitting in the first-order spectral region ($1000\text{--}2000\text{ cm}^{-1}$), in which the peak at $\approx 1580\text{ cm}^{-1}$ representing a perfect graphitic lattice and D1 band featured with the vibration mode of amorphous carbon, respectively.^[44] The peak intensity ratio ($I_{\text{D1}}/I_{\text{G}}$), which determines the degree of disorder, for pristine LIG can be estimated to be ≈ 0.73 (Table S1,

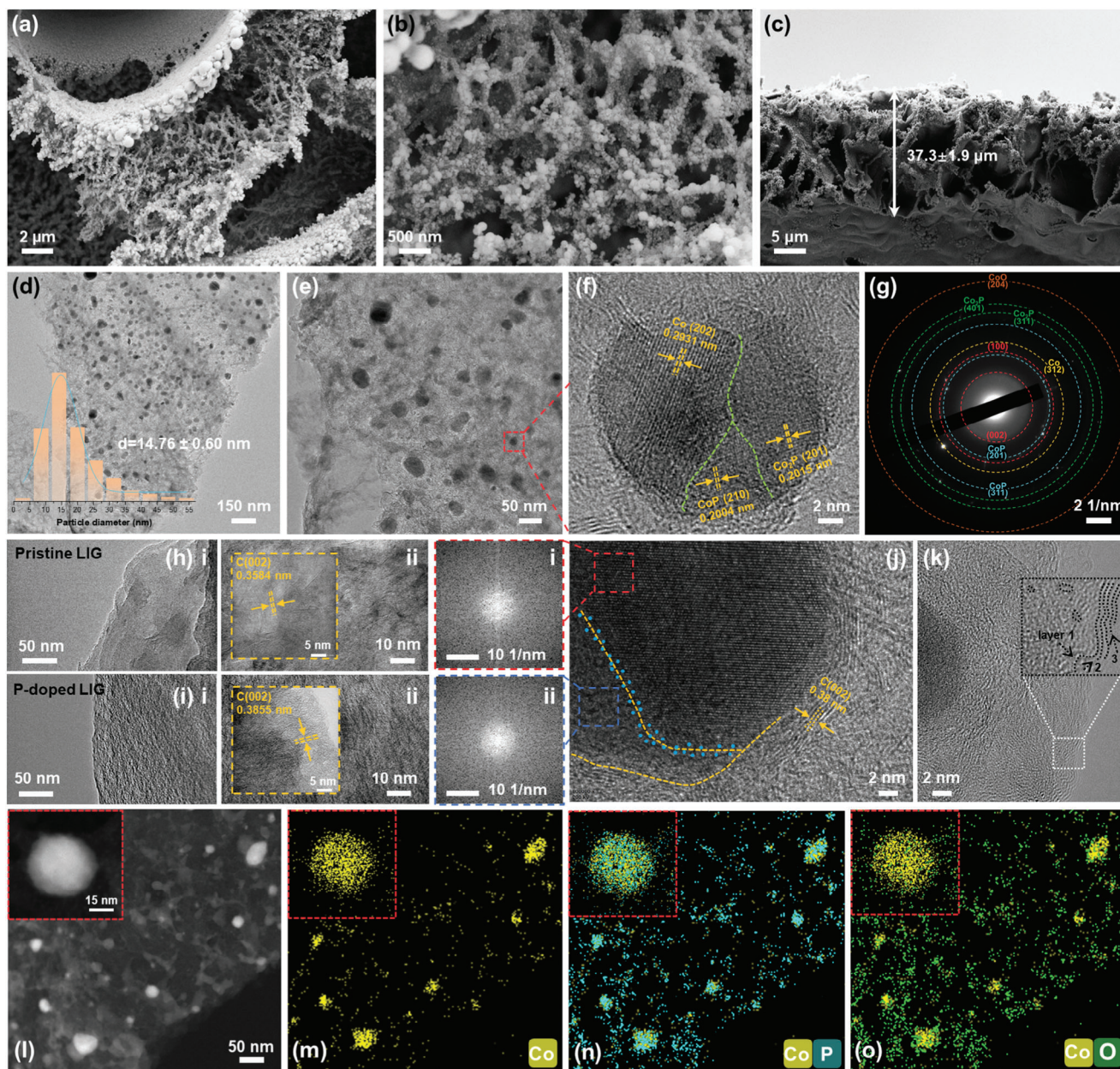


Figure 2. Nanoscale structural characterizations of $\text{Co}(\text{O}_x/\text{P}_x)$ -derived P-doped LIG. a) SEM, b) amplified SEM, and c) cross-sectional SEM images of $\text{Co}(\text{O}_x/\text{P}_x)$ @P-LIG. d,e) TEM images of corresponding core-shell nanostructures with scale bars: 150 and 50 nm, the inset of (d) shows the particle size distribution. f) HRTEM image of CoP_x core encapsulated by few-layered LIG. g) SEAD patterns of $\text{Co}(\text{O}_x/\text{P}_x)$ @P-LIG composites. h,i) HRTEM images of pristine LIG and P-doped LIG, the insets displaying the lattice fringe from orange area, respectively. j) HRTEM image of laser-induced core structures with crystalline/amorphous heterophases, the insets displaying the corresponding FFT patterns of the selected regions highlighted by the red and blue areas, respectively. k) HRTEM image of P-doped microporous LIG shell structure. l-o) HAADF-STEM images and the corresponding elemental maps of $\text{Co}(\text{O}_x/\text{P}_x)$ @P-LIG composites, the insets displaying the magnified tomography (scale bar: 15 nm), along with respective cobalt, overlapped cobalt-phosphorus and cobalt-oxygen elemental mappings.

Supporting Information). The introduction of P or Co in LIGs disrupts the perfect hexagonal sp^2 graphene lattice by inducing structural defects, indicated by higher of $I_{\text{D1}}/I_{\text{G}}$ values (≈ 1.09 and ≈ 1.14). The doping-induced increase in local electron density leads to a redshift of the entire spectra. However, in the laser-induced co-doping of P and Co, the ratio paradoxically decreases regardless of the atomic fraction within the composites

(Figure S8, Supporting Information). The reduced $I_{\text{D1}}/I_{\text{G}}$ ratio can be attributed to the formation of cobalt phosphide or oxide nanoparticles, subsequently reducing the vacancies in P-LIG. Thus, the resultant core-shell structure could potentially stabilize the doping state and mitigating the defects. The formation of $\text{Co}(\text{O}_x/\text{P}_x)$ nanoparticles and the potential for defect remediation may diminish the original defect sites, and while themselves

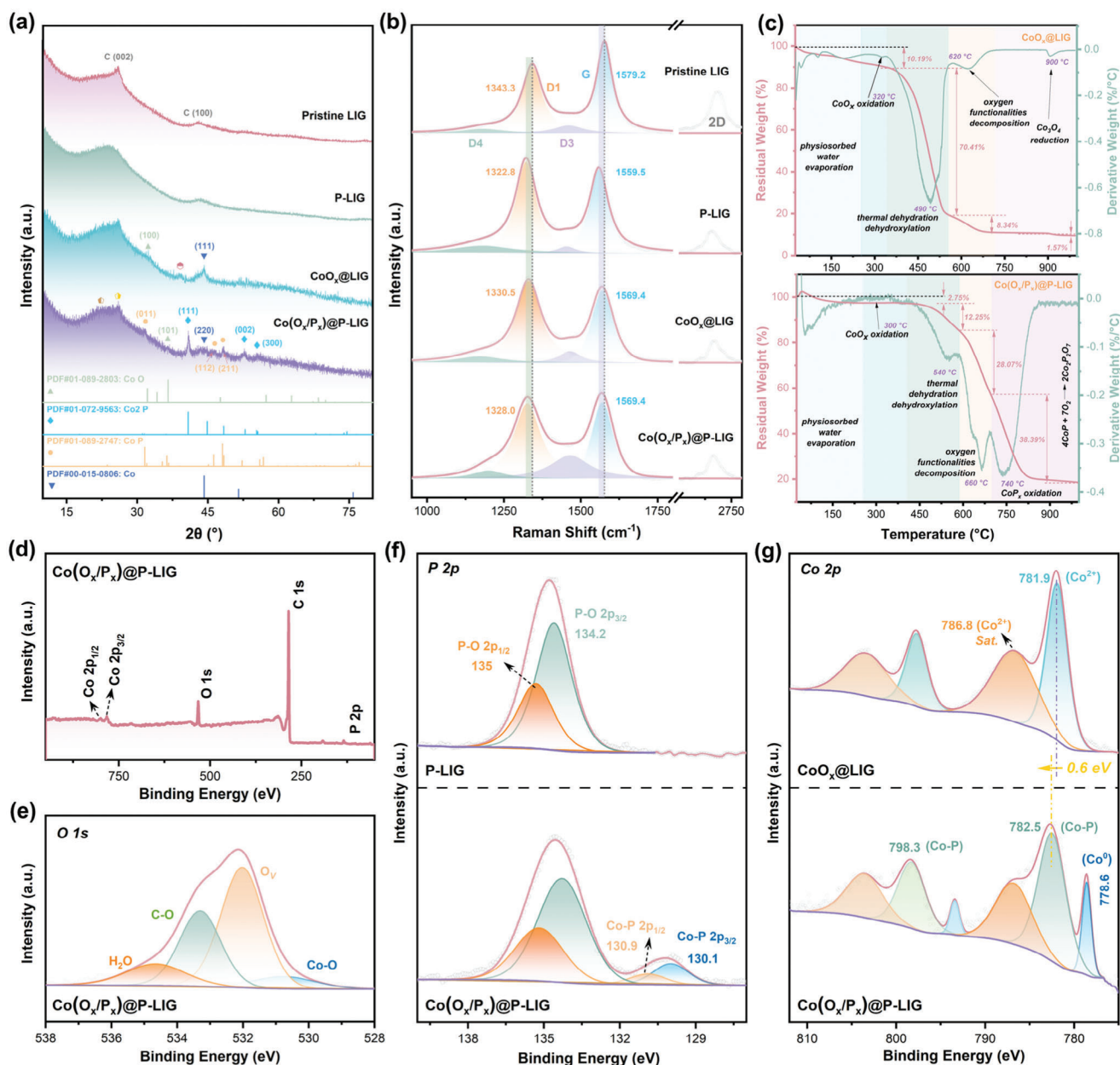


Figure 3. In-depth physicochemical and structural identification of nanoscale $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ active electrode material. a) XRD patterns and b) Raman spectra of pristine LIG, P-LIG, $\text{CoO}_x\text{@LIG}$, and $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$, respectively. c) TGA-DTG profiles of $\text{CoO}_x\text{@LIG}$ and $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$. d) XPS spectra of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$. e–g) High-resolution XPS spectra of O 1s, P 2p, and Co 2p for Co- and P-derived LIG nanocomposites.

could act as new active sites. Modulated charge distribution and improved structural stability may result in a more stable performance during redox reactions.

The $\text{Co}(\text{O}_x/\text{P}_x)$ content in the core was analyzed using thermogravimetric analysis (TGA) and derivative thermal gravimetry (DTG). The TGA-DTG curves of pristine LIG and P-LIG in air show four weight loss regions: i) physisorbed water evaporation (≈ 260 °C), ii) thermal dehydration and dehydroxylation (≈ 600 °C), iii) oxygen functionalities decomposition (≈ 800 °C), and iv) residual decomposition (to 1000 °C) (Figure S9a,b, Supporting Information). LIG exhibits superior thermal stability over

800 °C in an inert N_2 atmosphere,^[45] which supported by the fact that the majority of mass remains intact (Figure S9c,d, Supporting Information). Unlike the pyrolysis process associated with LIG, the doping of Co via oxidation at ≈ 300 °C and subsequent carbothermal reduction at ≈ 900 °C, preserve the cobalt as $\text{Co}(\text{II})$ or metallic Co and sustain the Co content at a steady value of $\approx 9.49\%$ (Figure 3c). In co-doped $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ composites, Co atoms undergo oxidation to form CoO_x and elevated thermal oxidation to form CoP_x at ≈ 740 °C,^[46] resulting in 18.54% $\text{Co}(\text{O}_x/\text{P}_x)$ within the active material (Note S2, Supporting Information).

The chemical bonding characteristics of $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ were further explored and compared with those of bare LIG, P-LIG, and $\text{CoO}_x@\text{LIG}$ using X-ray photoelectron spectroscopy (XPS). Peaks corresponding to C 1s, O 1s, P 2p, and Co 2p appear in the full XPS spectrum of $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$, indicating the coexistence of cobalt phosphides and oxides (Figure 3d). The weaker intensities of P 2p and Co 2p peaks compared to those of C 1s, O 1s peaks is consistent with previous findings, indicating core-shell configuration that LIG-shell may diminish the X-ray reflection emanating from the $\text{Co}(\text{O}_x/\text{P}_x)$ cores. The deconvoluted C 1s XPS spectrum exhibits four peaks at 284.8, 285.8, 287.6, and 290.6 eV corresponding to C—C/H, C—O—C, C—OH and C=O groups, respectively (Figures S10a and S11, Supporting Information).^[47] The high-resolution O 1s spectrum exhibiting four deconvoluted peaks could be identified in Co-derived nanomaterials compared with three for pristine or P-doped LIG (Figure 3e; Figure S10b, Supporting Information), and the O 1s peaks located at ≈ 530.4 eV can be attributed to the Co—O bonding. The characteristic peak at ≈ 532.0 eV results from CoO_x reduction and CoP_x oxidation, further confirming the generation of oxygen vacancies on the surface of $\text{CoO}_x@\text{LIG}$ and $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ (Figure S12, Supporting Information).^[48] The introduction of oxygen vacancies provides additional active sites, enhancing the pathways for electron/ion transport.^[49] Figure 3f shows the P 2p spectrum of P-LIG with deconvoluted peaks at 134.2 and 135.0 eV. The peak at ≈ 130.0 eV in the P 2p spectrum of $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ can be attributed to the amalgamation of phosphorus with metallic cobalt, indicating simultaneous doping of Co and P.^[50] In the Co 2p spectrum (Figure 3g), the peaks at 782.5 and 798.3 eV, accompanied by broad satellite peaks, can be attributed to divalent cobalt species, i.e., $\text{Co}(\text{II})$.^[51] The peaks related to the $\text{Co}(0)$ configuration of elemental cobalt in its reduced state within the decarbonized frameworks can be observed at 778.6 and 793.4 eV. Compared with $\text{CoO}_x@\text{LIG}$, a positive gap of 0.6 eV can be observed in the binding energy of $\text{Co } 2p_{3/2}$ XPS spectrum of $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$, indicating that the integration of P significantly impacts the chemical environment of Co atoms, which serving as electron acceptors to fill the unoccupied *d* orbital.^[52] Owing to the synergistic electron donation from P atoms and acceptance by Co atoms, the electronic structure and electron transfer capability of $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ are significantly improved.

To elucidate the mechanism behind the superior electron/ion transfer in $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$, Schottky contact, and *n-p* junction model have been established at the interface of metallic Co nanoparticles, semiconductive CoP_x cores, and *n*-type doped LIG phases to explore the synergistic effect within these three phases. The work functions of metallic Co, *p*-type CoP_x semiconductor,^[53,54] and *n*-type doped LIG are determined as ≈ 5.0 ,^[55] ≈ 5.2 ,^[56–58] and 4.6 eV,^[59] respectively (Figure 4a); the obtained work functions satisfy the requirements for forming a Mott-Schottky barrier and *n-p* junction. As the work function of CoP_x is higher than that of metallic Co, the conduction and valence bands of CoP_x bend downward to establish a Schottky contact (Figure 4b). Electrons spontaneously migrate from Co to CoP_x across the heterointerface until an equilibrium is achieved, resulting in electron accumulation on the CoP_x side and hole accumulation on the Co side. This electron redistribution facilitates the formation of a built-in electric field, which ef-

fectively gathers charged ions (EMIM^+ , BF_4^- , e^- , etc.) via the active sites. The increased electron density of CoP_x decreases the adsorption and/or desorption energy barriers, promoting electron/ion transfer within the active materials.^[56] Subsequently, a depletion layer forms along the heterointerface as electrons moving from *n*-type LIG toward CoP_x , achieving equilibrium at the Fermi energy level. To study the electron/ion transfer at the heterointerface between the active electrode materials and P.P, the valence band (VB)-XPS spectra were investigated and work functions of the Co/P-derived LIG nanocomposites were determined (Figure S13a,b, Supporting Information). The contact potential difference (ΔV), which alters the kinetic energy and finally changes the binding energy of the free electrons, can be calculated as follows: $\Delta V = \Phi - \phi$ (Φ and ϕ represent the work function of the materials and instrument, i.e., 4.6 eV, respectively).^[60] The work functions of $\text{CoO}_x@\text{LIG}$ and $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ are determined as 6.12 and 5.72 eV from the distance of the inflection point (IP) of VP-XPS curves, respectively. The bandgaps of both nanomaterials, as compared to those of *n*-type doped LIG composites, increase with the introduction of Co or P doping (Figure S13c,d, Supporting Information). The observed bandgap opening of P-LIG can be attributed to the presence of functional groups and defects (Note S3, Supporting Information).^[61]

Based on previous experimental results, the optimized electron/ion transfer effect can be properly predicted when doped LIG contacts a typical flexible conductive material, such as P.P (Figure S13e, Supporting Information). Since P.P owing a higher work function of ≈ 5.4 eV^[62] than P-LIG, the contact between them can result in a large work function difference, promoting the formation of a valley-type charge barrier. However, $\text{CoO}_x@\text{LIG}$ presenting a lower Fermi level facilitates the formation of a peak-type charge barrier at the heterointerface. The $\text{Co}(\text{O}_x/\text{P}_x)@\text{P-LIG}$ composite, which exhibits similar work function as P.P, can reduce potential charge barriers and enhance electron/ion transfer. Furthermore, the fundamental mechanism behind the mutual influence between the structure and electronic properties was investigated using density functional theory calculations;^[63,64] the corresponding optimized and adsorption configurations were shown in Figures S14 and S15 (Supporting Information). Compared with pristine LIG, the Co/P-derived electrode nanomaterials exhibit the enhanced electronic states near the Fermi level, thereby increasing the conductivity since the laser-induced heterostructures (Figure 4c; Figure S14b, Supporting Information). The deformation charge density further reveals the significant electron transfer, suggesting a potent interaction as evidenced by the absorption of IL ions (Figure 4d). This is similarly manifested in a substantial augmentation of adsorption energies, specifically, an elevation from -1.73 to -2.74 eV marking a 58.3% enhancement for EMIM^+ , and from -0.72 to -1.84 eV (155.5%) for BF_4^- (Figure 4e), which supports the distinguished ability to capture IL for high possibility of contacting with more active sites (Figure S16a, Supporting Information) that the IL droplet swiftly dispersed and absorbed with a considerable change rate of dynamic contact angle of 2.44°s^{-1} . As the doping within LIG, the change rates also significantly increase from 0.98°s^{-1} (bare LIG), to 1.92°s^{-1} (P-LIG) and 1.48°s^{-1} (Co-LIG).

The surface electrical conductivity of the active materials was investigated for determining the electron transfer resistance. The observed improvement from 2.15 (pristine LIG)

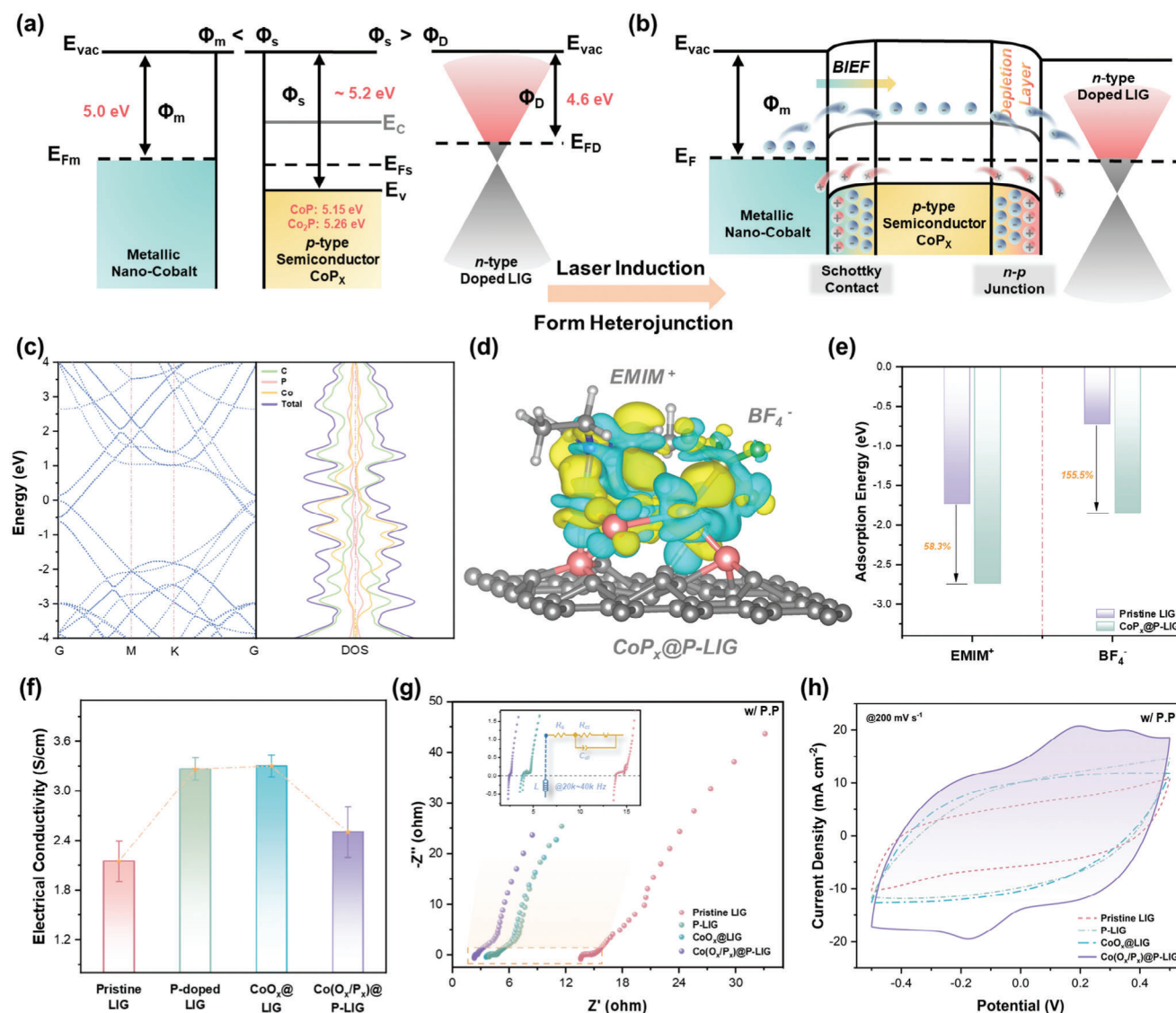


Figure 4. Schematic illustration of energy band structures a) before and b) after contact of $\text{CoP}_x@P\text{-LIG}$ heterojunction, where E_{vac} represents vacuum energy, Φ is vacuum electrostatic potential, E_F stands for Fermi level, E_C and E_V are conduction band and valence band, respectively. c) Band structures and DOS, d) deformation charge density of $\text{CoP}_x@P\text{-LIG}$ and as absorbed with sub-nanoscale ions (EMIM^+ , BF_4^-), respectively. e) The adsorption energy of different ions adsorbed on pristine LIG and $\text{CoP}_x@P\text{-LIG}$, respectively. f) Surface electrical conductivity of pristine LIG and Co/P-derived LIG composites. (g) EIS of the actuators containing active materials with P.P from 100 mHz to 100 kHz, and the inset displaying the charge transfer resistances, alongside the corresponding equivalent electrical circuit model (Randles circuit). h) CV curves of the actuators at a scan rate of 200 mV s⁻¹ in the potential window of -0.5 to +0.5 V.

to 3.27 S cm⁻¹ (P-LIG) results from the additional free electrons coming from the P atoms (Figure 4f). As the introduction of Co doping source, the conductivity of $\text{CoO}_x@P\text{-LIG}$ remains almost unchanged (≈ 3.30 S cm⁻¹), which is unaffected by the presence of cobalt oxides, owing to the reduction of elemental Co during laser induction. However, the conductivity of $\text{Co(O}_x/\text{P}_x)\text{@P-LIG}$ decreases expectedly to 2.50 S cm⁻¹, this can be attributed to the phosphides of cobalt that are further oxidized to form semiconductive CoP_x at elevated temperatures while reducing the content of reduced Co, as supported by the XRD results. The charge storage ability was evaluated using electrochemical cyclic voltammetry (CV) analysis in nonaqueous electrolytes (0.5 M EMIM-

$\text{BF}_4/\text{CH}_3\text{CN}$) under various scan rates (10–100 mV s⁻¹), and the ionic specific capacitance of $\text{Co(O}_x/\text{P}_x)\text{@P-LIG}$ reaches 116.0 F g⁻¹ at a scan rate of 10 mV s⁻¹ (Figure S16b, Supporting Information). During the DLW process, the high-temperature carbothermal environment promotes the formation of highly conductive metallic Co to a certain extent. This reduces the production yield of transition metal phosphides/oxides that exhibit pseudocapacitive properties, resulting in moderate overall specific capacitance performance (Figure S16c, Supporting Information). Nevertheless, the generation of metallic Co further improves the conductivity of the doped LIGs, conferring substantial advantages in terms of conductivity and fabrication simplicity over the

metal–organic frameworks commonly employed in supercapacitors (Figure S16d, Supporting Information).

Figure 4g shows the electrochemical impedance spectroscopy (EIS) results of the actuator, where the slopes of the straight line for $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ are slightly larger than those of the other composites, signifying a reduced Warburg impedance (Z_w) at the electrode–electrolyte interface, thereby favoring rapid ion diffusion and insertion in low-frequency region. The charge-transfer resistance (R_{ct}), responsible for the presence of high-frequency semicircle, cannot be directly observed for the Co/P co-doped sample, while significantly increase for the other doped LIG composites, implying the superior wettability of the surface electrodes and lower interfacial resistance of the P-doped electron-rich LIG. The superior Z_w and R_{ct} of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ nanosheets with high conductivity and capacitance can be attributed to improved ion diffusion and accelerated charge transfer in the redox reactions.^[65] Evidently, $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ with the addition of P.P (denoted as $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$) presents the redox reaction via a conventional two-electrode system, showing the reversible oxidation and reduction and thus significantly extending the conductive potential window (Figure 4h). As expected, the region encompassed by the CV curves substantially exceeded that of the other actuators featured EDLC behaviors, and at all scan rates, $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuators exhibited the highest areal capacitances (Figure S16e, Supporting Information). Such actuators demonstrate an impressive specific capacitance of 153.0 mF cm^{-2} at 100 mV s^{-1} , maintaining its rate capability even when the scan rate was doubled to 200 mV s^{-1} , achieving a capacity of 126.0 mF cm^{-2} . Notably, the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuators preserve a robust electrochemical charge capacity across scan rates varying from 10 to 200 mV s^{-1} . This indicates the exceptional actuation capabilities of such actuator can be attributed to the redox properties of $\text{Co}(\text{O}_x/\text{P}_x)$ nanoparticles that promote electron/ion transfer across the electrode–electrolyte interface and also within the EMIM- BF_4 embedded ion-exchange membrane (Figure S16f, Supporting Information).

2.2. Actuation Performances of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ Electro-Ionic Muscles

Owing to the superior electron/ion transfer and ion storage/diffusion capabilities, the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuators demonstrated an impressive bending actuation performance under ultralow potential inputs at variable frequencies in ambient conditions. The peak-to-peak bending displacement markedly increased to 2.04 times (14.05 mm), substantially exceeded the pristine LIG-based actuator (6.90 mm) under a square wave input of $\pm 0.5 \text{ V}$ at 0.1 Hz (Figure 5a). The bending actuation for sine-wave inputs is shown in Figure 5b; the largest deflection remains ascribed to the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator, reaching to 13.08 mm for peak-to-peak patterns. Remarkably, such actuators are capable of achieving a precise actuation of $\approx 280 \mu\text{m}$ under an exceptionally low voltage of $\pm 10 \text{ mV}$ (Figure 5c), and a subtler yet consistent actuation response still can be observed even at $\pm 2.5 \text{ mV}$ (Figure S17, Supporting Information), suggesting that even at minute electric fields, Co/P-derived LIG still exhibits electron/ion transfer and diffusion with high efficiency. Figure 5d indicates the improved tip bending performance under a range of

sinusoidal potential inputs from 0.1 to 0.5 V at 0.1 Hz compared with that of other actuators. The $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator consistently exhibits greater deflection regardless of the voltage variations. The actuation performance was also evaluated at excitation frequencies ranging from 0.1 to 15 Hz (Figure 5e), evidently, with the increase in oscillation frequency of input signals, the peak-to-peak displacement tends to be significantly decreased because of the limited ion-intercalation process.

Real-time electrical stimulation of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator was recorded under a sinusoidal potential input with frequencies varying from 0.1 to 10 Hz (Movie S1, Supporting Information). The bending amplitudes of all samples increased significantly at frequencies in the range of 6 – 10 Hz , implying fast structural resonance in response to high-frequency voltage. The response time plays a crucial role in determining the efficacy of an actuator in producing a significant displacement over a short period. The rise times observed at an input of 0.5 V direct current (DC) input are shown in Figure 5f, and the superior deflection measuring 9.2 mm was executed by the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator, highlighted by the minimal response duration of 1.38 s , which is ≈ 20 times less than pristine LIG actuator (3.5 mm) of a response time of 28.96 s . The long-term actuation durability of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator under $\pm 0.5 \text{ V}$ at 10.0 Hz was also assessed; the actuator exhibits minimal degradation ($<10\%$) even after $\approx 106\,000$ cycles under continuous electrical stimulation (Figure 5g; Note S4, Supporting Information). The comparative study of bending performances of the existing state-of-the-art electro-ionic soft actuators with active material-optimized electrodes (Figure 5h; Table S2, Supporting Information) demonstrates that the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator exhibits a competitive bending strain of $\approx 0.73\%$ with rapid response property under 0.5 V at 0.1 Hz .

In this work, motivated by the exceptional electro-ionic actuation performance of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator, an ingeniously designed soft robotic maneuver providing dual-mode stimulation for inducing differentiation and proliferation of MSCs is developed for the first time. MSCs are the most promising seed cells with inherent multipotency, which enables them to differentiate into various mesenchymal tissues, including bone, cartilage, muscle, and adipose tissues.^[66] As compared to the limitations associated with the procurement and proliferative abilities of fully differentiated cells, the multipotent capacity of MSCs highlights their importance in above tissue engineering.^[67] Differentiation and proliferation of MSCs can be influenced by external stimuli via signal transduction to induce cellular responses, which involves the application of mechanical or electrical stimulation to MSCs via biochemical signals.^[68] To overcome the bulky size and complexities associated with traditional cell stimulation apparatus,^[69] a novel cellular stimulation apparatus is developed (Figure 6a). The apparatus consisted of an automated culture platform designed using poly(ϵ -caprolactone) (PCL) aligned fiber scaffolds (Figure S18, Supporting Information) for facilitating differentiation of MSCs to form skeletal bone or muscle tissues.^[70] This platform (Figure S19, Supporting Information) is capable of providing mechanical strain for higher modulus (Table S3, Supporting Information) of the developed soft actuator to reach sufficient stiffness, as well as electrical stimuli originated from the conductive electrodes to MSCs, demonstrating a simplified structure, compact size, and biocompatibility.^[71]

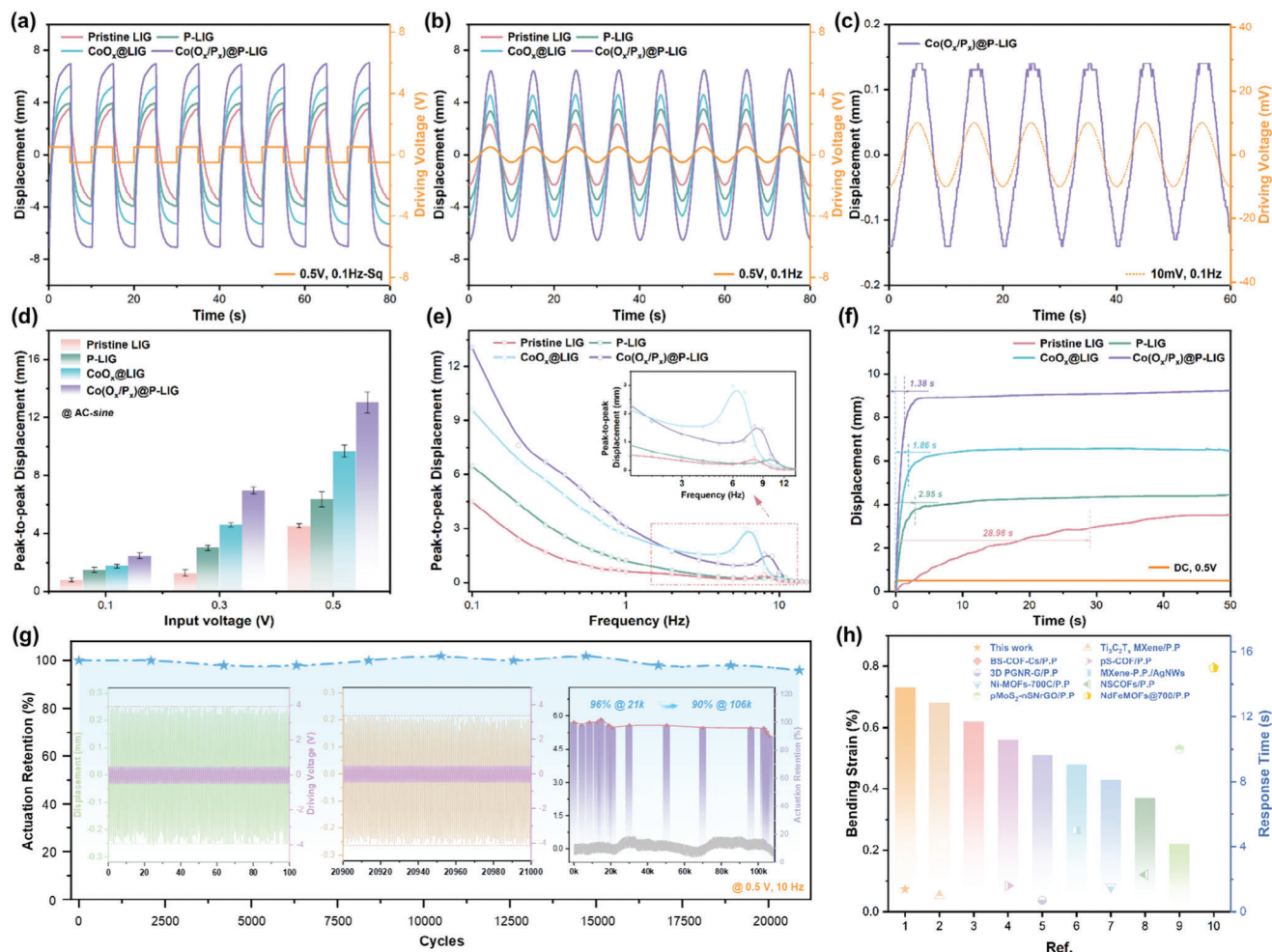


Figure 5. Electro-ionic actuation performances of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ soft actuators. Time-dependent comparative bending deflection of four LIG-based soft actuators with P.P under ± 0.5 V a) square wave and b) sinusoidal input potential at 0.1 Hz. c) Bending displacement of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ soft actuator under ± 10 mV sinusoidal potential input at 0.1 Hz. d) Comparative peak-to-peak displacements of the actuators under a range of input potentials from 0.1 to 0.5 V at 0.1 Hz. e) Comparative frequency responses from 0.1 to 15 Hz under sinusoidal input of ± 0.5 V. f) Comparative response time of the soft actuators under 0.5 V DC input voltage. g) Actuation retention of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ soft actuator under ± 0.5 V at 10.0 Hz, the insets representing the initial 1000 cycles (green), middle 1000 cycles at around 20 k cycles (brown), bending durability and retention of entire 106 k cycles, respectively. h) Comparative bending strains and response times of the developed soft actuator with previous similar works involving electrode-engineered electro-ionic actuators under ultralow voltage.

(Note S5, Supporting Information) without producing heat or shock (Figure S20; Note S6, Supporting Information), offering a competitive alternative for in vitro culture and differentiation of stem cells.

Following the application of a square-wave input signal to the actuator, the flexible electrodes of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator transmit the imposed electrical potential. Subsequently, grounding the distal end of the scaffold permits the delivery of homogeneous and directed electrical excitation. Under the combined load of the PCL scaffold and its own gravitational force, the device exhibits a deflection of 480 μm at 0.5 Hz and 201 μm at 2.0 Hz (Figure 6b; Movie S2, Supporting Information). The platform can impart a maximum magnitude of ≈ 1100 microstrains to a fibrous membrane at 0.5 Hz under a limiting voltage of 2.0 V alternating current (AC), satisfying the requisite stress stimuli for differentiation and proliferation of MSCs into osteoblasts^[72] and

myoblasts.^[73] Upon application of an external stimulus of one cycle of voltage input, the actuator enables the scaffold to undergo repeated cycles of expansion and contraction, achieving the transition from zero to maximum strain four times, i.e., 0.5 Hz for the actuator to 2.0 Hz for the scaffold (Figure 6c). Under continuous actuation for 10 min, the platform demonstrates an exceptional stability in its overall operational performance, exhibiting superior actuation durability (Figure 6d,e). The bending durability duration was further extended to ≈ 12000 s; the results demonstrate that the actuation retention remains above 90% for the initial 4000 s (Figure S21, Supporting Information). Although the actuation response began to fluctuate subsequently because of inevitable stress relaxation, the actuation response consistently follows the driving voltage, preserving a virtually constant bending displacement waveform. Notably, the central displacement shift of the peak-to-peak pattern also significantly reduces due to the

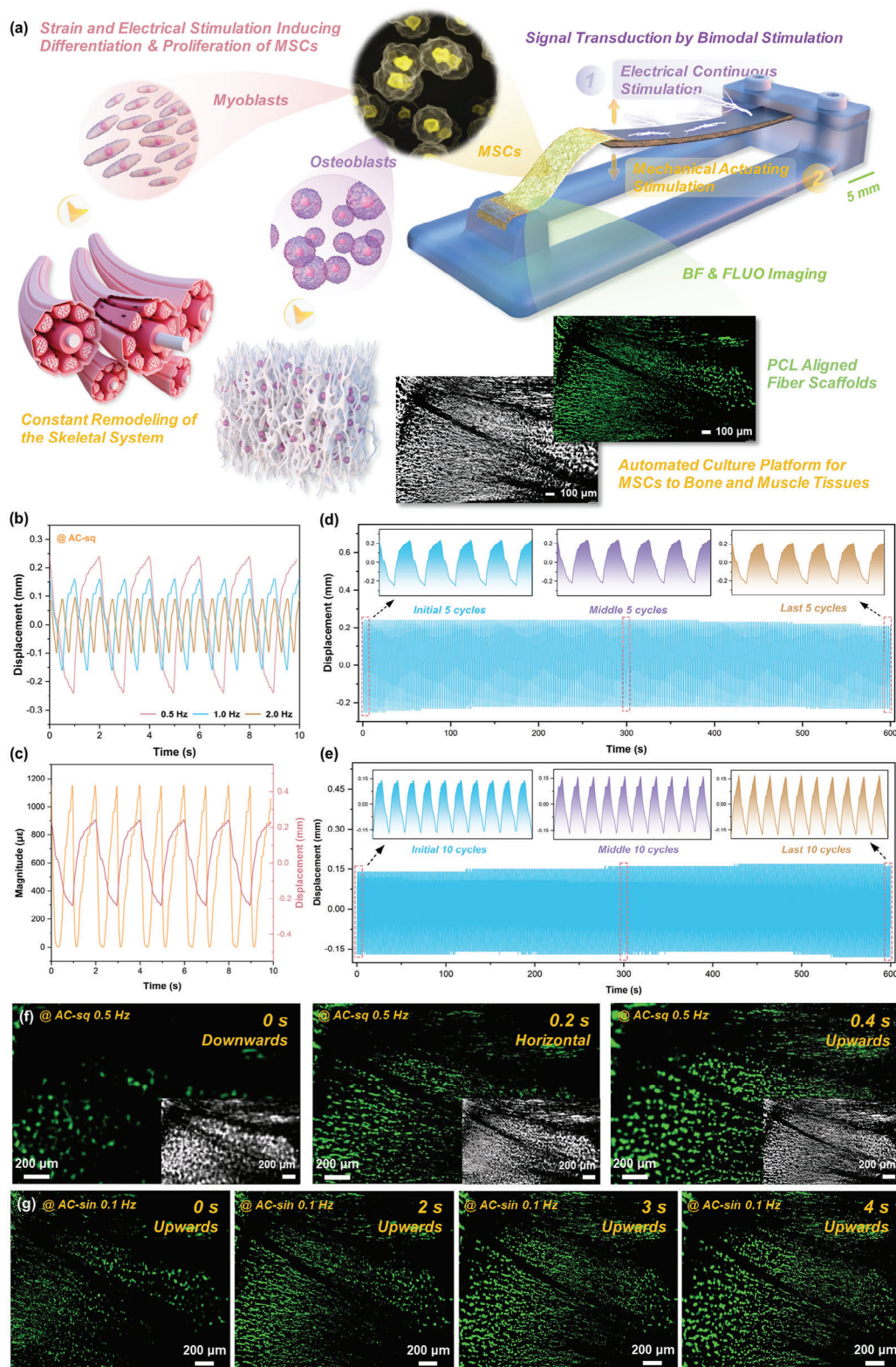


Figure 6. Potential application demonstrations of dual-modal stimulation inducing differentiation and proliferation of MSCs on PCL aligned fiber scaffolds via $\text{Co}(\text{O}_x/\text{P}_x)$ @P-LIG-P.P soft actuator. a) Schematic illustrations for the structure representation of MSCs automated culture platform. b) The actuation performance and c) applied strain magnitude (0.5 Hz) of the culture platform under the 2.0 V AC potential input of square wave at various frequencies of 0.5, 1.0, and 2.0 Hz, respectively. d,e) The actuation durability of the platform under 2.0 V AC of square wave at 0.5 and 1.0 Hz, respectively. f,g) Real-time BF&FLUO imaging of PCL scaffolds stretching with $\text{Co}(\text{O}_x/\text{P}_x)$ -derived soft actuator under square and sinusoidal potential inputs, respectively.

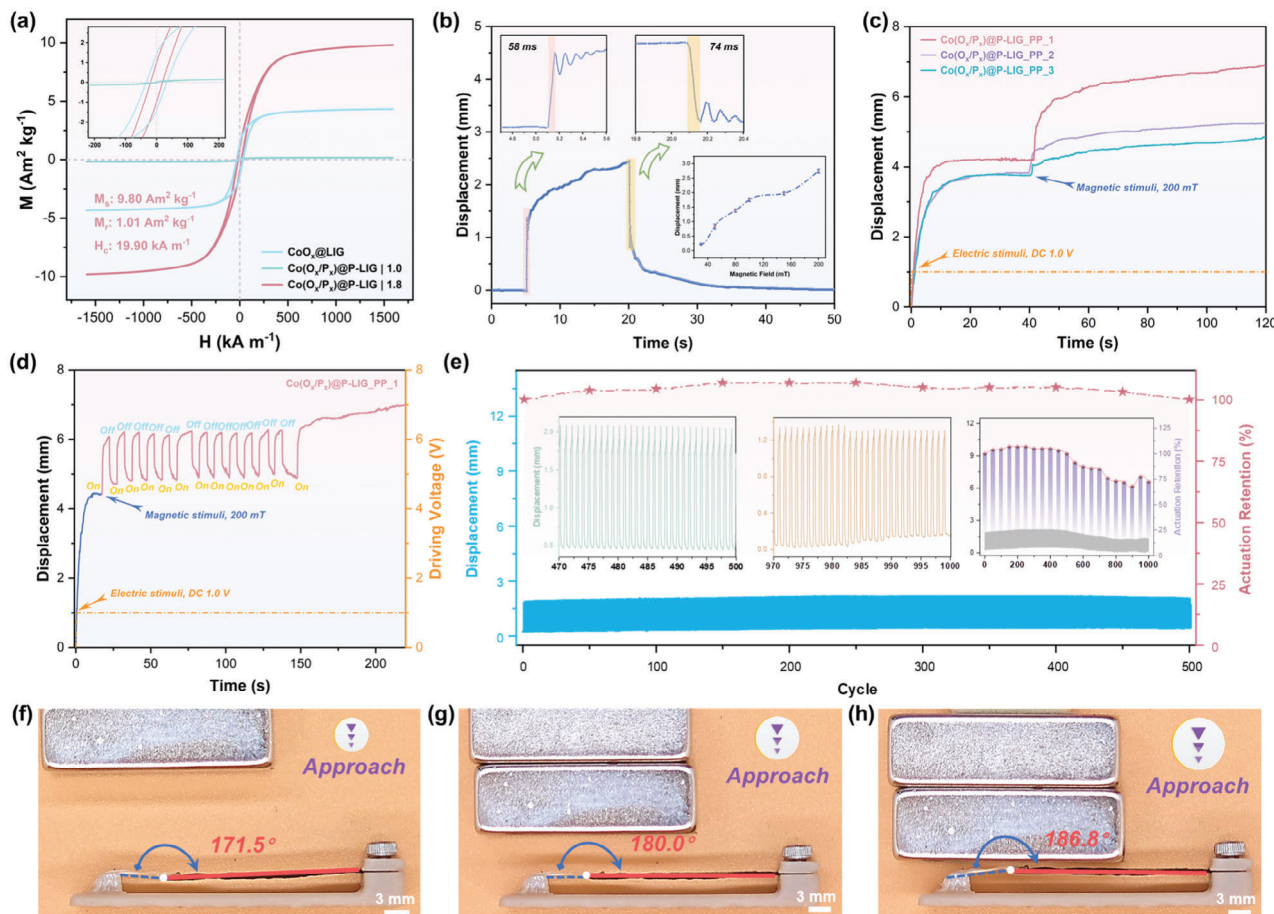


Figure 7. Dual-responsive actuation performances of the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ soft actuator under electric and magnetic fields. a) Magnetic hysteresis properties of nanoscale $\text{CoO}_x\text{@LIG}$ and $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ (Co/P atomic fraction: 1.0 and 1.8) at ambient temperature. Inset: the enlarged view of the low-field hysteresis loops with detailed magnetic properties in terms of saturated magnetization, remanence, and coercivity. b) Response time of the soft actuator with magnetic stimuli, and inset displaying the maximum deflection under various intensities of magnetic fields. c) Dual-modal actuation displacements of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ -based actuators initially under 1.0 V DC potential input then applying the magnetic field of 200 mT. d) Temporary deflection behavior of the actuators in response to instant on-off switching by magnetic stimuli while under ultralow electric stimuli. e) The actuation durability of the actuator stimulated by the unilaterally magnetic stimulation switching over 1000 cycles, the insets representing middle (green) and last (brown) 30 cycles, bending stability performance, and actuation retention, respectively. f–h) Visual representations depicting magnetic actuation of PCL aligned fiber scaffolds via the magnetoactive $\text{Co}(\text{O}_x/\text{P}_x)$ -derived soft actuator as the permanent magnet gradually approaches.

traction load from the PCL scaffolds. Owing to the stress recovery, the platform maintains its initial actuation performance even after 12 h (Note S4, Supporting Information). To demonstrate the ability of the platform for real-time imaging of cells subjected to stretching, analysis was conducted by adhering fluorescent microspheres to the PCL fiber scaffold, which observed by inverted fluorescence microscopy, driven by $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG-P.P}$ actuator. As illustrated in Figure 6f and Movie S3 (Supporting Information), under the voltage input of a 2.0 V at 0.5 Hz of square wave potential for 0.5 s, the device underwent 2 instances of strain alteration, namely, a transition from the low position (downward) to a horizontal position, subsequently continuing its deformation to reach the apical high position (upward), the detailed working principles are provided in Note S7 (Supporting Information). Under excitation by an identical sinusoidal voltage wave, the PCL scaffold moves upward from a horizontal position to the high position, with notable contraction occurring between the fiber interstices, providing an effective platform for differentiation and

proliferation of MSCs (Figure 6g; Movie S4, Supporting Information).

The elemental forms of transition metals, as well as their oxides or phosphides, possess soft ferromagnetic properties, paving the way for an innovative transition from traditional electrochemical actuation to a highly efficient and controllable electromagnetic dual-mode actuation approach. The visual representation in Figure S22a (Supporting Information) illustrates the magnetic arrangement of $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ nanocomposites subjected to an external permanent magnet, presenting the possible arrangement of dipoles in the applied magnetic field. Magnetic hysteresis analysis shows that the $\text{Co}(\text{O}_x/\text{P}_x)\text{@P-LIG}$ nanocomposite with a Co/P atomic fraction of ≈ 2 exhibits the largest saturated magnetization (M_s) of 9.80 $\text{Am}^2 \text{kg}^{-1}$, a low remanence (M_r) of 1.01 $\text{Am}^2 \text{kg}^{-1}$ and minimal coercivity (H_c) of 19.90 kA m^{-1} at room temperature (Figure 7a). The large value of M_s can be attributed to the presence of Co_2P and metallic Co having exceptional magnetic properties, as supported by XRD results.

Similar properties are observed for CoO_x/P nanomaterials owing to the presence of highly crystalline elemental Co. The equal atomic ratio for Co/P of $\text{Co}(\text{O}_x/\text{P}_x)/\text{P}$ -LIG nanocomposite exhibits negligible magnetic properties, primarily due to the formation of CoP, a compound with semiconducting properties that contributes negligibly to magnetism. Additionally, the absence of magnetic transition metal phases prevents the formation of a strong magnetic phase, reducing the overall magnetic properties by disrupting the inherent magnetic structure.^[74] The small M_r and H_c values observed in the optimized $\text{Co}(\text{O}_x/\text{P}_x)/\text{P}$ -LIG indicates that such composites can magnetize and demagnetize under magnetic stimuli, facilitating precise and effortless control by applying magnetic fields (Figure S22b, Supporting Information).

Furthermore, the $\text{Co}(\text{O}_x/\text{P}_x)/\text{P}$ -LIG actuator exhibits the rapid response (58 ms) and release time (74 ms) w/o the introduction of external magnetic field (Figure 7b), revealing results that are even faster than the majority of soft actuators fabricated from ferromagnetic materials (Figure S23, Supporting Information). The displacement of the actuator maintains a quasi-linear correlation with the magnitude of the magnetic field (inset to Figure 7b), indicating a high functional responsiveness. Figure 7c shows the results of the dual responsiveness of the developed actuator assessed via both electric and magnetic stimuli. Initially, under 1 V DC, the actuation displacement reaches a value of ≈ 4 mm; then, the DC displacement nearly doubles under the influence of a 200 mT magnetic field, demonstrating significantly improved actuation performance. In modulating magnetic field, the developed actuator is shown to swiftly magnetize and demagnetize (Figure 7d), demonstrating fast on-off switching capabilities that can be utilized in complex actuation systems, such as soft robotics. The actuation durability of the actuator has also been investigated which stimulated using the unilateral magnetic stimulation for 1000 switching cycles, and the retention during the initial 600 cycles consistently remains above 90% (Figure 7e). However, mechanical fatigue develops in the actuator owing to sustained unilateral oscillations, results in reduced performance retention, and such issue can be mitigated by further adjusting the direction and frequency of the applied magnetic field. Notably, the actuation response waveform remains almost unchanged even after 1000 cycles. Based on the magnetic characteristics of the active electrode materials, an attempt was made to induce strain and deformation within scaffolds by applying external magnetic fields. By pre-adjusting the position of the scaffold to be downward, the unidirectional approach of the permanent magnet was utilized to induce biaxial strain in the platform. Upon gradually increasing the magnetic field to 200 mT, the scaffold undergoes significant displacement deformation and changes its position from downward to upward, i.e., from an obtuse angle (171.5°) to a reflective angle (186.8°) (Figure 7f–h; Movie S5, Supporting Information), achieving a maximum strain change of 85 microstrains.

The demonstration of electric and magnetic actuation shed more light on the possibility of facilitating active and passive driving mechanisms for the MSCs culture platform, thereby holding promise for the further simplification of its system complexity through the use of alternating electric or magnetic fields. By thinking the large-scale production to apply the dual-responsive actuated PCL platform to in vitro applications, there are still

certain challenges for the future in elucidating mechanisms, large-scale deployment, and managing environmental conditions that need to be addressed to unlock the full potential of such soft actuators for biomedical applications. Although the actuation response and characteristics of the proposed actuator has been extensively evaluated using electrical, magnetic, chemical, and mechanical analysis methods, the complex transmission patterns of ion and electron intercalation and deintercalation at the electrode, detailed in Note S8 (Supporting Information), are yet to be completely understood. Thus, systematic studies on micro-actuation mechanisms are necessary to establish a theoretical foundation for understanding macroscale actuation behaviors. Further, the construction of functional modules by combining multiple PCL platforms to facilitate large-scale production of scaffolds is required. To further improve the productivity of PCL platforms, increasing the level of automation within the fabricated modules and integrating advanced flexible sensors for real-time monitoring and analysis is essential. Lastly, temperature and humidity fluctuations can affect the actuator response (Note S9, Supporting Information), necessitating further attention to heat and humidity management to promote cell viability and ensure smooth operation of the PCL platforms.

3. Conclusion

In summary, magnetic-responsive electro-ionic soft actuators based on novel Co/P co-doped nanoparticles with core-shell configurations incorporated into 3D graphene frameworks via single-step laser direct patterning have been developed. The physicochemical properties, electrochemical characteristics, and energy-band structures of the integrated $\text{Co}(\text{O}_x/\text{P}_x)/\text{P}$ -LIG nanocomposites were thoroughly investigated to understand the underlying mechanisms for designing high-performance multi-responsive actuators. Structurally, the designed nanofiber-based heterointerface across electrodes and electrolytes significantly enhances high-efficient electron/ion transfer within the designed actuator. The active material exhibited a narrow magnetic hysteresis with saturated magnetization of $9.80 \text{ Am}^2 \text{ kg}^{-1}$ accompanied small remanence and coercivity values. The $\text{Co}(\text{O}_x/\text{P}_x)/\text{P}$ -LIG-P actuators exhibited a high areal capacitance of 153.0 mF cm^{-2} , resulting in significant improvement of actuation performance with a peak-to-peak displacement of 13.08 mm under an ultra-low $\pm 0.5 \text{ V}$ and doubled DC deflection under 200 mT, ultrafast response time of 1.38 s with exceptional stability ($<10\%$ degradation even after $\approx 106\,000$ cycles), detectable bending of $\approx 280 \mu\text{m}$ even under $\pm 10 \text{ mV}$. The promising demonstration of the soft actuator-supported automated platform, capable of providing mechanical strain and electrical stimuli simultaneously for the differentiation and proliferation of MSCs, shows great potential of advancing biomedical soft robotics in the near future.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

artificial muscle, automated culture platform, ionic actuators, laser-induced graphene, multi-responsiveness

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